Rutherford's Atomic Model Quiz

Physics • Grade 11

Student Name: _____ **Date:** _____ **Score:** ___ / 3

Read each item carefully and answer.

Part A: Choose the right answer

Tick one box.

1. Rutherford's model improved upon the plum pudding model by suggesting that most of the atom's mass and positive charge is concentrated in a small, dense region called the nucleus. Which of the following statements is correct?

- ☐ **A.** Rutherford's model proposed that the nucleus is composed of electrons.
- ☐ **B.** Rutherford's model introduced the concept of a nucleus containing most of the atom's mass and positive charge.
- ☐ **C.** The plum pudding model accurately described the structure of atoms.
- ☐ **D.** The plum pudding model suggested a uniform distribution of positive and negative charges.

Part B: True or false

Tick True or False.

2. The scattering of alpha particles in Rutherford's experiment indicated that atoms are mostly empty space, with a small, dense nucleus.

☐ **True** ☐ **False**

Part C: Fill in the blank

Choose a word from the word bank.

3. The nuclear model of the atom, proposed by Rutherford, laid the foundation for understanding atomic structure and led to the development of _____ mechanics.

Word bank: **gravitational electromagnetic classical quantum**

Answer Key

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1. D. The plum pudding model suggested a uniform distribution of positive and negative charges. — Rutherford's model was a significant improvement over the plum pudding model, which suggested a uniform distribution of positive and negative charges.

2. True — The scattering of alpha particles in Rutherford's experiment indicated that atoms are mostly empty space, with a small, dense nucleus.

3. quantum — The nuclear model of the atom, proposed by Rutherford, laid the foundation for understanding atomic structure and led to the development of quantum mechanics.